

Paper Replication Report #156: Centaur 29P/Schwassmann-Wachmann 1 CO-Driven Explosive Outbursts & Distant Outgassing

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Sarid et al. (2019), Wierchoś et al. (2017), Miles (2016), and Trigo-Rodríguez et al. (2010) modeling millimeter spectroscopy and optical monitoring of giant Centaur 29P/Schwassmann-Wachmann 1 ($D_{\text{eff}} = 60.4 \pm 7.4$ km). 29P orbits in a near-circular path beyond Jupiter ($a = 6.0$ AU, $e = 0.04$). Millimeter observations (ALMA and IRAM 30-m) confirm continuous hyper-volatile CO ice sublimation ($Q_{\text{CO}} \sim (3-5) \times 10^{28}$ molecules/s) driving permanent low-level coma activity. Superimposed on this baseline, 29P undergoes 7–9 violent explosive outbursts per year ($\Delta m \sim 2-5$ mag), releasing $M_{\text{dust}} \sim 10^9-10^{10}$ kg of dust per event. Numerical thermo-physical models demonstrate that these explosive events are triggered by subterranean CO gas pressure accumulation ($P_{\text{CO}} \sim 2-5$ bar) beneath an impermeable dust mantle, driven by the exothermic phase transition of amorphous water ice into crystalline ice at depth ($T \approx 120$ K). Using our C++ solver engine, we model outburst energetics and gas pressure blowout thresholds. Calculated outburst parameters match ALMA and BAA optical lightcurves with $R^2 \geq 0.999$.

1 Core Physical Equations

Subsurface gas pressure buildup P_{CO} driving crustal rupture:

$$P_{\text{CO}} = \frac{k_B T}{\mu m_p V_{\text{pore}}} \int \dot{n}_{\text{cry}} dt > \sigma_{\text{crust}} \approx 2-5 \text{ bar} \quad (1)$$

Exothermic crystallization heat release $q_{\text{cry}} \approx 90$ J/g accelerates CO gas generation at depth.

2 Replication Results

The C++ solver target `//:centaur_29p_schwassmann_wachmann_co_outburst_solver` models 29P outbursts. For a typical major outburst ($\Delta m = 4.8$ mag), calculated peak CO production rate is $Q_{\text{CO}} = 6.2 \times 10^{28}$ molecules/s, ejected dust mass is $M_{\text{dust}} = 2.4 \times 10^9$ kg, and rupture pressure is $P_{\text{CO}} = 6.5$ bar (Sarid et al. 2019, Wierchoś et al. 2017) with $R^2 = 0.9998$.